

# HTML TABLES REFERENCE MANUAL

## Mastering Rowspan, Colspan & Structured Data Layouts (No CSS Style)

### Introduction to Tabular Structures

HTML tables are used exclusively to present structured, grid-based information such as schedules, marksheets, lists, or transactional data. **Important Rule:** Modern web development practices prohibit using HTML tables to design page layouts. Tables must only be utilized for tabular data structures.

### Core HTML Table Elements

| HTML Tag | Semantic Purpose                                                                | Code Structure Syntax Example   |
|----------|---------------------------------------------------------------------------------|---------------------------------|
| <table>  | Defines the parent container block wrapping the entire table grid structure.    | <table border="1"> ... </table> |
| <tr>     | Defines an individual horizontal Table Row within the grid.                     | <tr> ... </tr>                  |
| <th>     | Defines a Table Header cell. Bolded and centered horizontally by default.       | <th>Header Title</th>           |
| <td>     | Defines a standard Table Data cell. Regular weight and left-aligned by default. | <td>Cell Data</td>              |

### Traditional Layout Attributes (CSS Excluded)

| Attribute   | Functional Parameter Purpose                                                                            | Direct Attribute Syntax Example |
|-------------|---------------------------------------------------------------------------------------------------------|---------------------------------|
| border      | Sets the visual boundary outline around the table and internal grid cells.                              | border="1"                      |
| cellpadding | Defines the internal spacing distance (in pixels) between cell wall boundaries and its content text.    | cellpadding="10"                |
| cellspacing | Defines the spacing distance (in pixels) separating individual cells from each other.                   | cellspacing="0"                 |
| bgcolor     | Applies a background fill color to the table, a row, or a single cell using hexadecimal or color names. | bgcolor="#F4F6F7"               |

## Standard Table Code Implementation

```
<table border="1" cellpadding="8" cellspacing="0">
  <tr>
    <th>Student Name</th>
    <th>Subject</th>
    <th>Marks</th>
  </tr>
  <tr>
    <td>Aniket</td>
    <td>HTML Basics</td>
    <td>95</td>
  </tr>
</table>
```

# Spanning Dimensions: Colspan & Rowspan

## 1. Column Spanning (Horizontal Merging)

To stretch a table cell horizontally across multiple columns, use the colspan attribute inside a <th> or <td> element. The value specifies how many column widths the cell will merge into.

☐☐ **The Displaced Cell Rule:** When you write colspan="N" on a cell, that single cell consumes the space of N cells in that row. You **MUST completely delete (omit)** the subsequent N-1 <td> tags in that row. Failure to do so will push the remaining cells outward, creating an uneven and broken grid boundary.

```
<!-- Row with cell merging across two columns -->
<tr>
  <td colspan="2"><b>Total Score</b></td>
  <td>183</td> <!-- This is the remaining 3rd column cell -->
</tr>
```

## 2. Row Spanning (Vertical Merging)

To stretch a table cell vertically across multiple rows, use the rowspan attribute. The value specifies how many rows deep the cell will expand downward.

☐☐ **The Downward Displacement Rule:** When you write rowspan="N" on a cell in Row 1, that cell physically occupies space in the subsequent rows below it (Row 2, Row 3, etc.). Therefore, when coding those lower rows, you **MUST completely omit** the cell matching that vertical column position. The browser already knows that the cell from Row 1 is occupying that slot.

```
<table border="1" cellpadding="8" cellspacing="0">
  <tr>
    <td rowspan="2" bgcolor="#EAF1F1">Aniket</td> <!-- Starts spanning -->
    <td>Monday</td>
  </tr>
  <tr>
    <!-- Rowspan occupies the first spot! Omit Name TD cell completely here -->
    <td>Tuesday</td>
  </tr>
</table>
```

## 3. Lecture Practical Lab Assignment

Create a file named cinema.html in VS Code and construct the following structural grid using only raw HTML table elements, traditional border attributes, and spanning properties (strictly **no CSS**):

| Movie Title Column (Vertical)                  | Showtime Column | Screen Column |
|------------------------------------------------|-----------------|---------------|
| Marvel's Avengers<br>(Spans 2 rows vertically) | 2:00 PM         | Screen A      |
| [Spanned and Omitted in HTML!]                 | 6:00 PM         | Screen A      |
| Spider-Man (Single Row)                        | 4:00 PM         | Screen B      |